From Point Colorimetry to Governed Spatial–Spectral Quality Assessment

A Systematic Framework for Appearance Quality Control of Heterogeneous Industrial Materials

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Publication Date: 16 January 2026

Abstract

Industrial colour and appearance quality control has traditionally relied on point-based spectrophotometric measurements or subjective visual inspection. While point measurements provide high spectral precision, they inherently assume material homogeneity and fail to represent spatial variability. Visual inspection, although sensitive to spatial context, lacks reproducibility, traceability, and auditability.

This whitepaper presents a systematic, high-level framework for the assessment of colour and appearance quality in heterogeneous industrial materials, based on spatial imaging rather than isolated point measurements. The framework integrates visible colour imaging (VIS), near-infrared imaging (NIR), and multispectral imaging under controlled and versioned illumination conditions, and evaluates quality using a hierarchical pixel-to-component-to-batch model. Central to the framework is a recipe-driven governance layer that embeds perceptual impact weighting, traceability, and reproducible decision logic across sites and systems.

The contribution of this work lies not in the invention of new sensing technologies, but in the systematic operational integration of spatial–spectral measurement, controlled illumination, perceptual decision modelling, and governance into a coherent quality-control paradigm suitable for industrial deployment.

1. Introduction

Across many industrial sectors—including powders, granulates, food ingredients, coatings, polymers, recycled materials, and composite surfaces—visual appearance constitutes a critical quality attribute. Despite advances in sensing and automation, quality control in these domains remains largely dependent on point-based colour measurements, often supplemented by human visual inspection.

Point spectrophotometry is highly effective for homogeneous samples but becomes unreliable when colour and appearance vary spatially. Localised particles, inclusions, agglomerates, stains, or texture variations can dominate perceived quality while being insufficiently captured by point sampling. Visual inspection compensates for this limitation but introduces observer dependency, lighting variability, and limited traceability.

The resulting discrepancy between instrumental measurement and perceived quality leads to inconsistent decisions, rework, and disputes across the supply chain. Addressing this discrepancy requires a fundamental shift in how colour and appearance quality are conceptualised and measured.

2. Conceptual Shift: From Point Measurement to Spatial Quality Models

The framework described here replaces point-based evaluation with a spatial statistical model of quality. The fundamental unit of measurement is the pixel, which contains visible colour information and, where available, near-infrared and multispectral responses. Pixels are analysed in spatial context rather than isolation.

Spatially related pixels are grouped into components, representing visually meaningful entities such as particles, inclusions, stains, or clusters of similar appearance. Quality assessment is performed hierarchically at the pixel, component, and batch levels. This enables the use of distributions, proportions, and outlier analysis rather than single averaged values.

By treating quality as a spatial and statistical construct, the framework aligns more closely with human visual perception and with practical industrial acceptance criteria for heterogeneous materials.

3. Integrated Multimodal Imaging

Visible imaging forms the basis for perceptual colour assessment, with image data mapped to standardised colour spaces such as CIE XYZ and CIELAB following calibration. Rather than relying on a single colour value, colour distributions across the sample are analysed.

Near-infrared imaging is incorporated as an orthogonal information layer. NIR response is sensitive to material composition and absorption behaviour and can reveal differences between matrix material and contaminants even when visible colour contrast is limited.

Multispectral imaging using a limited number of defined spectral bands extends discriminatory capability beyond RGB without requiring full hyperspectral resolution. The intent is robustness and practical applicability in quality-control environments rather than chemical identification.

The framework does not claim novelty in these modalities individually. Its contribution lies in their coordinated use as a single, integrated decision framework.

4. Controlled Illumination as a System Parameter

A defining feature of the framework is the explicit treatment of illumination as an integral system parameter. Spectral power distribution, intensity, and spatial directionality of illumination are defined, recorded, and versioned as part of the measurement process.

Controlled illumination ensures reproducibility across time and locations and enables meaningful comparison between measurements. Sequential illumination from different directions, with fixed camera geometry, provides additional information on relative surface structure and texture. These indicators are reproducible and sufficient for quality assessment without requiring full three-dimensional metrology.

5. Detection, Clustering, and Component Statistics

Deviations are detected using combined criteria incorporating perceptual colour distance, spectral distance, spatial coherence, and near-infrared response. Detected pixels are clustered into components, which are quantified in terms of count, size distribution, surface fraction, and internal statistics.

While spatial component statistics are well established in prior art, their role here is explicitly enabling rather than novel. Their significance lies in how they are embedded within a governed, perceptually weighted decision framework rather than in the statistics themselves.

6. Appearance, Texture, and Relative Structure

Appearance quality extends beyond colour to include texture and surface structure. Optical texture metrics derived from spatial image statistics complement colour and spectral information. Relative surface structure can be inferred through directional illumination analysis, providing reproducible indicators of roughness, agglomeration, or surface irregularity relevant for industrial quality control.

7. Perceptual Impact Weighting and Decision Logic

A distinguishing element of the framework is the explicit incorporation of perceptual impact weighting. Deviations are not evaluated solely by fixed thresholds but can be weighted according to size, contrast, and spatial distribution.

This approach enables quality decisions that reflect visual and functional relevance while remaining deterministic, explainable, and auditable. Decision rules are explicitly defined and reproducible.

8. Recipe-Driven Governance and Traceability

All measurements are executed according to **versioned recipes** defining sample preparation, illumination conditions, calibration references, analysis rules, and acceptance criteria. Recipes are centrally governed and version-controlled, enabling consistent application across multiple stations and sites.

This governance layer—encompassing traceability, auditability, and fleet-level consistency—is a central element of the framework and a key point of differentiation from existing imaging or spectroscopic systems.

9. Artificial Intelligence Within a Governed Framework

Artificial intelligence may be integrated for segmentation, classification, anomaly detection, or decision support. AI models may contribute directly to acceptance decisions, provided they are explainable, confidence-scored, version-locked, and embedded within the same recipe-driven governance structure as deterministic rules.

AI is therefore neither excluded nor unconstrained; it operates within explicit boundaries that preserve reproducibility, auditability, and regulatory suitability.

10. Related Prior Art and Distinctions

Spatial and spectral imaging techniques for quality control are well documented in prior art, including hyperspectral imaging for process analytical technology and spatial analysis in food and material inspection. Image-based colour measurement and pixel-level prediction are also established.

However, existing disclosures do not describe a framework that **systematically integrates** spatial statistics, combined VIS–NIR–multispectral imaging, controlled and versioned illumination, perceptual impact weighting, and recipe-driven governance into a single operational quality-control paradigm. The present framework addresses this integration gap.

11. Reference Implementation (Non-Limiting)

A reference implementation may combine visible imaging, near-infrared imaging, and multispectral acquisition under controlled multi-directional illumination, followed by spatial segmentation, component analysis, and impact-weighted decision modelling. This example is provided solely to demonstrate technical feasibility and does not limit the scope of the framework.

12. Methodological Scope and IP Positioning

The protected scope of this publication lies in the **system-level integration** of spatial measurement, multimodal imaging, controlled illumination, perceptual decision modelling, and governance. Any implementation that applies these principles in combination—independent of specific hardware, algorithms, or architectures—falls conceptually within the framework described herein.

13. Conclusion

This whitepaper defines a systematic framework for appearance quality control of heterogeneous industrial materials. By redefining quality as a spatial and statistical construct and embedding measurement within a governed, perceptually informed workflow, the framework bridges the gap between human visual inspection and instrument-based quality control. Its value lies not in individual technologies, but in the coherence, reproducibility, and operational integrity of the integrated system.

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